Phase 3: Software Requirements Specification
for
Intelligent AI Powered Circular Campus Resource Exchange and Asset Life Cycle Management System

Version 1.2

Prepared by Group 6

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COSC 336: Introduction to Software Engineering

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Revision History

Name	Date	Reason for Changes	Version
Group 6	4th October 2026	Initial Phase 3 requirements document.	1.0
Group 6	11th October 2026	Aligned template sections; clarified feature scope, quantity allocation, custody, return and end-of-life outcomes; completed interface requirements and reporting definitions.	1.1
Group 6	15th October 2026	Revised Sections 1 to 6 for readability and traceability: tabulated the requirement prefixes, product scope and planning assumptions; mapped product functions to Section 4; labeled the privacy and data-delivery requirements; and aligned requirement formatting document-wide.	1.2

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1 Introduction

1.1 Purpose

This SRS defines the Group 6 COSC 336 semester prototype, building on Phase 1 v1.0 and Phase 2 v1.1. It specifies user roles, external interfaces, functional and nonfunctional requirements, the five AI functions, sustainability measures, and operational workflows. Architecture and detailed design are deferred to Phases 4 and 5.

1.2 Document Conventions

Requirement identifiers are stable and are used throughout Sections 3 to 6 and the appendices.

Prefix	Meaning
FR-<FEATURE>-nnn	Functional requirement
NFR-<CATEGORY>-nnn	Nonfunctional requirement (performance, safety, security, privacy, quality, AI explainability)
UI-nnn	User interface requirement (Section 3.1)
SI-nnn	Software interface requirement (Section 3.3)
CI-nnn	Communications interface requirement (Section 3.4)
AI-REQ-nnn	AI requirement applying across features
SUS-nnn	Sustainability measure
DD-nnn	Data and delivery requirement (Section 6.3)
BR-nnn	Business rule (Section 5.5)
AD-nnn	Assumption or dependency (Section 2.7)
UC-nn	Use case (Appendix B.4)
TBD-nnn	Open decision (Appendix C)

“Shall” states required behavior. Every requirement carries either an Acceptance or a Verification clause.

Requirements are labeled Must or Should. Must requirements belong to the semester prototype. The three Should refinements are combined approvals (FR-TRANSFER-002), escalation routing (FR-TRANSFER-008), and a purchase indicator (FR-REQUEST-008).

High and Medium indicate development order, not inclusion. All five AI functions remain required. Future enhancements are outside scope.

Assumptions in this document are prototype planning assumptions. They do not establish institutional policy; delegation, retention and exception handling remain open. Open decisions are TBD-001 (technology stack), TBD-002 (emission factors), TBD-003 (life-extension method), TBD-004 (matching and scoring factors), TBD-005 (AI response times) and TBD-006 (delegation and retention).

1.3 Intended Audience and Reading Suggestions

The specification is written for four audiences.

Audience	Read first	Use for
Stakeholders	Sections 1–2, Section 5.5, Appendix B	Scope, roles, business rules, workflows
Developers	Sections 3–6	Interfaces, requirements, AI behavior
Testers	Requirement identifiers and their checks; Appendix B	Success, alternative and exception paths
Instructors and academic reviewers	Section 1, Appendices B and C	Assessment of scope, traceability and open decisions

1.4 Product Scope

The platform connects departmental needs with available campus assets so that resources are reused rather than replaced. It maintains asset history through inspection, maintenance, repair, refurbishment and end-of-life decisions. Conventional workflows are extended by AI classification, semantic matching, sustainable-action advice, a conversational assistant and report summaries, with human approval and fallback to conventional workflows when AI is unavailable.

Product scope
Included in the prototype
Accounts, roles and department management
Asset registration and publication
Search, requests, reservations, approvals and transfers
Inspection, maintenance, repair, refurbishment and end-of-life records
The five AI functions
Notifications, audit records, and operational and sustainability reports
Excluded from the prototype
Complete procurement, accounting, transport and warehouse-management systems
Live integration with university systems
Physical work, which staff perform and record (Phase 2, Section 1.2)

The prototype uses synthetic data, so measured campus impact would require a real deployment. Possible later extensions are institutional or identity integration, and demand forecasting.

1.5 References

Reference	Version or date	Use
Group 6 Phase 1 Initial Plan and Requirement Gathering Document	v1.0, 13 September 2026	Baseline scope, stakeholders and numbered requirements.
Group 6 Phase 2 Feasibility Study	v1.1, 26 September 2026	Prototype boundaries, operating assumptions, technology proposal and risks.

Reference	Version or date	Use
COSC 336 Project Description (Project-September2026.pdf)	Fall 2026	Project outcomes and phase deliverables.
Phase 3 Requirements Document Slides (Phase_3_Slides.pptx)	Supplied lab deck	Requirements quality, use cases and Phase 3 structure.
Circular Campus Project and Feasibility Study lab decks	Supplied lab decks	Project context and feasibility guidance.
Software Requirements Specification.dotx	Supplied course template	Required section and subsection structure.

The specification relies on the following course sources. Where the project brief and the lab slides give different milestone dates, the confirmed instructor schedule governs. This specification does not establish a submission deadline.

2 Overall Description

2.1 Product Perspective

The prototype is a new, self-contained university application that brings departmental asset management into one system. It complements institutional procedures and does not integrate with institutional registers, identity, procurement or accounting systems. The implementation stack is confirmed in Phase 4 (TBD-001).

2.2 Product Functions

The prototype provides the following functions, each specified in Section 4.

Function	Specified in
User and Access Management	Section 4.1
Asset Registration and Publication	Section 4.2
Resource Search and Filtering	Section 4.3
Resource Requests and Reservations	Section 4.4
Approval and Transfer Management	Section 4.5
Inspection, Maintenance, Repair and Life-Cycle Tracking	Section 4.6
AI-Assisted Asset Classification	Section 4.7
AI Resource Matching, Ranking and Explanations	Section 4.8
Sustainable-Action Recommendations	Section 4.9
Natural Language Assistant	Section 4.10
Reporting, Notifications and Auditability	Section 4.11

2.3 User Classes and Characteristics

Roles follow Phase 1 Section 3.1 and Phase 2 Section 2.1. Requesters and representatives are frequent users; the remaining roles perform specialist tasks. A person may hold several assigned roles. Permissions that are not listed for a role are denied. Every signed-in role can use the assistant within its existing access, and technical administration grants no business authority. Procurement, Finance and Sustainability Officers share a single actor in Figure 1 but retain distinct report scopes. Final institutional delegation remains TBD-006.

Role and characteristics	Permitted work	Report access
Requester Staff with a departmental need.	Search; submit/track own requests; review matches; express interest; report defects on accessible assets.	Own request/asset status; no general financial or operational reports.
Department Representative / Asset Custodian Authorized departmental asset manager.	Register/edit/publish own assets; submit requests; approve outgoing reservations/release; confirm incoming receipt; report defects; propose end-of-life actions.	Own department operational and sustainability reports and relevant costs.
Business Administrator Internal reuse process owner.	Resolve escalated discrepancies and authorize end-of-life actions within delegation. Transfer intervention is for escalations.	Cross-department operational and sustainability reports and relevant exception costs.
Maintenance Staff Inspection and service specialist.	Inspect and record service, repair/refurbishment, costs/outcomes and assessed condition.	Maintenance activity and associated costs.
Facilities Staff Movement and storage staff.	Coordinate collection, delivery/storage and record physical handovers.	Assigned transfer arrangements and handover records; no general reports.
Procurement Officer Purchasing specialist.	Review unmet needs and reuse/availability before purchases.	Demand, availability, reuse and avoided-purchase reports.
Finance Officer Financial reviewer.	Review values, costs and estimated savings.	Financial reports and supporting relevant activity records.
Sustainability Officer Environmental performance reviewer.	Review reuse, repair, diversion, emissions and next-action advice; prepare summaries.	Sustainability reports and their calculation inputs.
System Administrator Technical operator.	Manage accounts, roles, departments and technical configuration. Business authority requires a separate role.	Account and technical audit records; business reports require a separate assigned role.
AI Services (external system) External AI service. Supports the AI functions; cannot authorize transactions.	Provides classification, matching, sustainable-action advice, assistant responses and report summaries.	None; AI output is validated against recorded activity.

AI Services supports the five AI functions and cannot authorize transactions. Procurement/Finance/Sustainability share UC-12 diagram participation but retain distinct report scopes. Final institutional delegation is TBD-006.

2.4 Operating Environment

Users access the prototype through campus computer browsers. It requires authentication, persistent records, file storage and an external AI service. AI credentials remain server-side. Browser, version and hosting details are settled in Phase 4. The figures below are Phase 2 planning assumptions, not measured university data (Phase 2, Section 1.3).

Phase 2 planning assumption	Value
Participating departments and service units	20
Authorized users	200
Daily active users	40
Concurrent sessions	5
Asset records	5,000
Available listings	500

2.5 Design and Implementation Constraints

The four-person team works within course deadlines and a limited service budget, using synthetic data. Hosting is on GitHub, and the team works with the instructors and lab engineers, committing regularly and attributably. Transactions respect university policy, delegated authority and privacy. AI outages or usage limits must not remove conventional workflows. The programming language for Phases 4 to 6 is open: the Phase 3 lab wording names Python, Java or C++, while Phase 2 proposes a web stack (TBD-001).

2.6 User Documentation

User documentation accompanies the prototype as short written guides or in-application help; packaging is selected during design. Guides cover publishing assets, searching and submitting requests, approving reservations and transfers, and movement and receipt. They also cover discrepancies, maintenance and end-of-life, reviewing AI suggestions, and reading report estimates. Setup, configuration, data loading, backup and restore, and operation while AI is unavailable (NFR-QUAL-005) are also covered.

2.7 Assumptions and Dependencies

ID	Assumption or dependency	Effect if unavailable
AD-001	Authorized staff keep asset availability and condition records current (Phase 1, Sections 3.4 and 7.4).	Listings and matches become unreliable.
AD-002	The external AI service is available within its usage limits (Phase 2, Section 4.3).	AI features are unavailable; conventional search, forms, and approvals continue.

ID	Assumption or dependency	Effect if unavailable
AD-003	The synthetic dataset is sufficient for functional and performance tests (Phase 2, Section 4.2).	Test results are less meaningful; no claims about real-world accuracy can be made.
AD-004	University staff perform physical collection, inspection, repair, and disposal (Phase 1, Section 2).	The system can record but not complete these activities.
AD-005	Each participating department has an authorized representative to approve releases and confirm receipt (Phase 2, Section 2.1).	Reservations and transfers cannot proceed.
AD-006	Published factors for sustainability estimates can be obtained (Phase 1, Section 6).	Affected estimates are shown as unavailable rather than calculated.

Quantity accounting. Every unit a requester asks for is in exactly one of three states: accepted fulfillment, actively allocated to a confirmed reservation, or unallocated outstanding and still eligible for a further reservation. For an active request, requested = accepted fulfillment + actively allocated quantity + unallocated outstanding quantity. These terms are defined in Appendix A. A request is not complete while any unallocated outstanding quantity remains, and unmet needs remain visible to the Procurement Officer.

Custody. Formal departmental custody is accountable assignment; physical holder and location track actual movement. Safety. Rejection or cancellation cannot restore stock recorded as unsafe. Section 4 and Appendix B apply these prototype rules, and institutional delegation and retention remain TBD-006.

3 External Interface Requirements

3.1 User Interfaces

UI-001 The system shall provide sign-in, role-appropriate navigation, asset registration and editing, and marketplace search. It shall provide areas for request and reservation status, approvals, transfers and receipts, maintenance records, and sustainable-action review. Assistant interaction and reporting complete the set. Forms shall show required fields, preserve invalid input and explain correction steps (FR-ASSET-005, FR-REQUEST-002, NFR-QUAL-002). **Verification:** complete each role's permitted workflows and confirm denied actions are unavailable and rejected on direct access.

UI-002 The system shall visibly distinguish confirmed asset data from AI suggestions, show processing/error states and uncertainty, and require user review before submitting assistant-prepared requests. Quantity and status views shall distinguish available, reserved, in-transit, received and unfit portions. **Verification:** review normal, partial, missing-data and AI-outage scenarios. Detailed screen mockups belong to Phase 4.

3.2 Hardware Interfaces

No specialized hardware interface is required for the prototype. Users use ordinary computers, keyboards and browser file upload for asset photographs and documents. Physical equipment, transport, weighing, scanning, sensors and automated inspection are outside scope; staff enter their observations and handover records. Verification: all prototype workflows can be completed using an ordinary browser without a dedicated device.

3.3 Software Interfaces

SI-001 The application shall exchange account/session data with its authentication service; asset, request, allocation, approval, transfer, maintenance and audit records with persistent storage; and asset files with file storage. The browser shall not receive service credentials. **Verification:** access, persistence and unauthorized-access tests (NFR-SEC-001 to 003, NFR-QUAL-003). No institutional-system interface is included.

SI-002 Server-side AI requests shall contain only task-required permitted information. Classification receives supplied asset metadata and files; matching receives request and listing data; action recommendations receive asset history, costs and demand; the assistant receives accessible records and policy; report summaries receive computed report data. Responses shall be treated as suggestions and validated before use. **Verification:** inspect request and response content, invalid-output handling and AI-disabled fallback. Phase 2 proposed React and TypeScript, Cloudflare hosting, Supabase and Gemini. Provider versions, APIs and final implementation choices are design decisions subject to TBD-001.

3.4 Communications Interfaces

CI-001 Browser/server and external-service traffic shall use encrypted connections. Protected operations shall require authentication and authorization; transfer or reservation confirmations shall not be duplicated by a retry, and failures shall not leave partly applied records (NFR-SEC-001/002, NFR-QUAL-003). **Verification:** configuration review, denied-access tests and interrupted/retried transaction tests.

CI-002 Notifications shall be in-application records with a permitted link and required action (FR-REPORT-006 to 008). Network or notification failure shall preserve saved business records and display an appropriate status. Email, SMS and live institutional messaging are not required prototype interfaces. **Verification:** simulate loss of connection and notification delivery failure.

4 System Features

These eleven features retain Phase 1 scope. Must/Should inclusion is defined in Section 1.2. Each feature follows the template subsections; Appendix B supplies detailed scenarios.

4.1 User and Access Management

4.1.1 Description and Priority

Priority: High. Manage departments, accounts, roles and protected access (Phase 1, Section 4.1).

4.1.2 Stimulus/Response Sequences

Administrator edits account/department → system validates → confirmation saves changes and audit history. Invalid credentials/actions are rejected. See UC-01.

4.1.3 Functional Requirements

FR-USER-001 [Must] The system shall allow a System Administrator to create and update department records. **Acceptance:** Administrator creates and edits a department; the change is visible to users.

FR-USER-002 [Must] The system shall allow a System Administrator to create, update, and deactivate user accounts and assign departmental membership and roles. **Acceptance:** A deactivated user cannot sign in; role changes take effect at next sign-in.

FR-USER-003 [Must] The system shall authenticate users before granting access to protected functions and allow them to sign out. Invalid credentials shall not establish a session. **Acceptance:** Invalid sign-in is rejected; a signed-out user cannot open protected pages.

FR-USER-004 [Must] The system shall restrict records and actions according to the user's assigned role and department (Section 2.3), including requests made through the assistant. **Acceptance:** Allowed and denied actions are tested for every role, through the interface and the assistant.

FR-USER-005 [Must] The system shall record changes to accounts, departmental membership, and permissions, identifying the responsible user, action, and time. **Acceptance:** Each change appears in the audit log with user, action, and time.

4.2 Asset Registration and Publication

4.2.1 Description and Priority

Priority: High. Record assets and publish eligible surplus using saved, validated metadata (Phase 1, Section 4.2).

4.2.2 Stimulus/Response Sequences

Representative enters details → reviews classification → saves confirmed metadata → publishes eligible quantity. Missing fields block publication; active allocations block withdrawal. See UC-02/03.

4.2.3 Functional Requirements

FR-ASSET-001 [Must] The system shall assign each asset record a unique identifier and store its name, description, category, quantity, owner, department, custodian, location, purchase date, estimated value, condition, availability, photographs, and supporting documents. **Acceptance:** A saved record shows all fields and a unique ID.

FR-ASSET-002 [Must] The system shall distinguish individually tracked assets from quantity-based batches and track batch quantities separately by condition, allocation and life-cycle status. A reservation or transfer shall affect only its allocated quantity. **Acceptance:** For a batch of 10 available units, reserving 3 leaves 7 available and 3 reserved. Cancellation restores the 3; dispatch affects only those 3. Status quantities sum to the recorded total.

FR-ASSET-003 [Must] The system shall allow an authorized representative to publish surplus, underutilized, or unwanted assets within their responsibility without central pre-approval. **Acceptance:** A representative publishes an asset and it appears in the marketplace.

FR-ASSET-004 [Must] The system shall reuse saved asset information when preparing a listing, allowing the user to correct or complete it without re-entering it. **Acceptance:** Listing form opens pre-filled from the asset record.

FR-ASSET-005 [Must] The system shall prevent publication when the asset name, a positive quantity, owner, department, custodian, location, or declared availability is missing or invalid, identifying the problem while preserving entered information. **Acceptance:** Each missing or invalid publication field blocks publication with a specific message; entered information is kept. Zero and negative quantities are rejected.

FR-ASSET-006 [Must] The system shall display each listing's recorded condition and availability and clearly mark unknown condition or incomplete information. Publication alone shall not indicate approval for transfer. **Acceptance:** Listing with unknown condition is visibly marked.

FR-ASSET-007 [Must] The system shall allow a listing to be withdrawn only when no active reservation or transfer applies to it. **Acceptance:** Withdrawal of a reserved listing is blocked.

FR-ASSET-008 [Must] The system shall record changes to asset details, identifying the responsible user, time, and information changed. **Acceptance:** An edit appears in the asset history with before and after values.

4.3 Resource Search and Filtering

4.3.1 Description and Priority

Priority: High. Provide conventional discovery and the non-AI search baseline (Phase 1, Section 4.3, Req. 13).

4.3.2 Stimulus/Response Sequences

User enters keywords/filters → system displays eligible listings and current stock. No results leave the need open; retired/disposed units are unavailable. See UC-05.

4.3.3 Functional Requirements

FR-SEARCH-001 [Must] The system shall provide keyword search over published assets. **Acceptance:** Searching a known keyword returns the matching listing.

FR-SEARCH-002 [Must] The system shall filter search results by category, condition, department, location, quantity, availability, asset age, estimated value, and life-cycle status. **Acceptance:** Each filter, alone and combined, returns only matching listings.

FR-SEARCH-003 [Must] When a search returns no results, the system shall say so without closing any related resource request. **Acceptance:** An empty search shows a message; the linked request stays open.

FR-SEARCH-004 [Must] The system shall not show retired or disposed assets as available. **Acceptance:** A disposed asset never appears in availability results.

4.4 Resource Requests and Reservations

4.4.1 Description and Priority

Priority: High. Record needs and reserve stock without over-allocation (Phase 1, Section 4.3).

4.4.2 Stimulus/Response Sequences

Requester submits need → expresses interest → supplier approves → system checks stock and remaining unallocated demand → confirms reservation. Partial fulfillment keeps unmet need active. See UC-04/07.

4.4.3 Functional Requirements

FR-REQUEST-001 [Must] The system shall allow requesters to submit a resource request specifying category, purpose, specifications, quantity, preferred condition, urgency, location, and required date. **Acceptance:** A complete request is saved with a unique ID and status.

FR-REQUEST-002 [Must] The system shall identify missing required information, non-positive quantities, or invalid dates in a request while preserving entered information. **Acceptance:** Each invalid input produces a specific message; data is kept.

FR-REQUEST-003 [Must] The system shall allow a requester to express interest in a listed asset and link it to an existing request without re-entering the request's details. An expression of interest shall not reserve the asset. **Acceptance:** Interest is linked to the request; available quantity is unchanged.

FR-REQUEST-004 [Must] The system shall confirm a reservation only after authorized approval and an atomic check of current available stock and the request's unallocated outstanding need. Confirmed reservations shall not exceed either quantity, including competing approvals. **Acceptance:** Two concurrent approvals for the last unit confirm only one reservation. A further reservation on a partially fulfilled request succeeds only within its remaining unallocated need.

FR-REQUEST-005 [Must] The system shall display request and reservation status, allocated and outstanding quantities, and the next required action and responsible role. **Acceptance:** After each step, the status page shows requested, accepted, actively allocated and unallocated outstanding quantities, the next action and responsible role.

FR-REQUEST-006 [Must] If a requester accepts partial fulfillment, the system shall keep the remaining need open unless the requester explicitly closes it. Further reservations shall be permitted while unallocated outstanding quantity remains, including in Reserved and Partially Fulfilled states. **Acceptance:** A request for 10 units receives 3 and can reserve the remaining 7; it cannot reserve an eighth additional unit. An explicit closure stops further matching and allocation.

FR-REQUEST-007 [Must] The system shall allow requests or reservations to be canceled before dispatch, releasing any reserved quantity. After dispatch, cancellation shall not return resources to available stock until custody and receipt are reconciled. **Acceptance:** Cancellation before dispatch restores quantity; after dispatch it does not.

FR-REQUEST-008 [Should] The system shall allow the requester to indicate whether the request would otherwise result in a purchase, for use in avoided-purchase reporting (SUS-001). **Acceptance:** The indicator is stored and used by the avoided-purchase report.

4.5 Approval and Transfer Management

4.5.1 Description and Priority

Priority: High. Track release, physical handover and accepted receipt (Phase 1, Section 4.4).

4.5.2 Stimulus/Response Sequences

Reservation → authorized release → dispatch with physical tracking → accepted receipt and formal reassignment. Discrepancy keeps allocation open; confirmed return reconciles condition and unmet need. See UC-08.

4.5.3 Functional Requirements

FR-TRANSFER-001 [Must] The system shall record approval or rejection of reservation, release, and transfer requests, identifying the authorized decision-maker, date, and reason. **Acceptance:** Each decision is stored with these details and shown in history.

FR-TRANSFER-002 [Should] The system shall allow related approvals to be completed together when the approver holds the required authority, recording each decision separately. **Acceptance:** A combined approval creates separate decision records.

FR-TRANSFER-003 [Must] The system shall maintain a transfer record linking the assets, approved quantities, supplying and receiving departments, collection or delivery arrangements, responsible personnel, destination, and supporting documents. **Acceptance:** Transfer record shows all linked information.

FR-TRANSFER-004 [Must] The system shall prevent release for ordinary use when required approvals are missing, mandatory suitability information is unresolved, or the asset is recorded as unsafe or unusable, and shall identify the issue. **Acceptance:** Release attempt in each of these cases is blocked with a message.

FR-TRANSFER-005 [Must] The system shall allow an authorized receiving representative to record received quantity, observed condition, acceptance or discrepancy, and actual physical holder and location. A discrepancy shall keep the transfer open without automatically changing formal departmental custody. **Acceptance:** A short or damaged delivery becomes Disputed; actual holder and location are recorded, formal assignment remains unchanged, and disputed units are unavailable.

FR-TRANSFER-006 [Must] The system shall complete a transfer only after accepted receipt of the approved quantity, then update authorized ownership, formal departmental custody, destination assignment, availability and request fulfillment together. Dispatch shall record physical holder and location without completing the transfer. A resolved return shall release its allocation only after the supplying representative confirms return receipt and condition; unsafe returned units shall remain unavailable. **Acceptance:** Dispatch updates physical tracking only. Accepted receipt completes the transfer and fulfillment together. A rejected receipt leaves it open. Confirmed return closes it as Returned, restores unmet need and updates the supplying asset state according to condition.

FR-TRANSFER-007 [Must] The system shall record handling and transport costs associated with each transfer. **Acceptance:** Transfer costs are stored and used in net financial savings (SUS-004).

FR-TRANSFER-008 [Should] The system shall allow unresolved exceptions, such as transfer discrepancies, to be referred to the Business Administrator for resolution (Phase 2, Section 2.1). **Acceptance:** A referred issue appears for the Business Administrator with its history.

4.6 Inspection, Maintenance, Repair and Life-Cycle Tracking

4.6.1 Description and Priority

Priority: High. Retain life-cycle history and assessed condition, including repair and end-of-life decisions (Phase 1, Section 4.5).

4.6.2 Stimulus/Response Sequences

User reports defect → maintenance inspect/service → authorized assessment restores service or records Unfit. End-of-life rejection restores the appropriate prior state. See UC-09/10.

4.6.3 Functional Requirements

FR-MAINT-001 [Must] The system shall maintain a chronological history of acquisition, allocation, utilization, inspection, maintenance, repair, transfer, refurbishment, donation, component recovery, recycling, retirement, and disposal events, identifying the date and responsible user. **Acceptance:** Each event type appears in the asset history in date order.

FR-MAINT-002 [Must] The system shall allow authorized users to report defects, request inspection or repair, and track the status of those requests against the asset. **Acceptance:** A defect report is created and its status can be followed.

FR-MAINT-003 [Must] The system shall allow maintenance staff to record inspection findings, maintenance, repair, or refurbishment actions, associated costs, outcomes, and the asset's condition after service. **Acceptance:** Completed work order shows findings, actions, costs, and new condition.

FR-MAINT-004 [Must] The system shall keep assets recorded as unfit for ordinary use unavailable for ordinary-use reservation until an authorized condition update establishes suitability. Returning an asset to service shall not override an existing reservation or transfer. Rejection of an end-of-life proposal shall restore the appropriate prior state, including Unfit or Retired, and shall not establish suitability for use. **Acceptance:** An Unfit asset remains Unfit after end-of-life rejection and cannot be reserved. Return to service requires a suitable condition assessment and preserves any active allocation.

FR-MAINT-005 [Must] The system shall require authorized approval before recording donation, component recovery, recycling, or disposal as completed, and retain the decision, reason, date, responsible personnel, and supporting records. **Acceptance:** Completion cannot be recorded without approval.

FR-MAINT-006 [Must] The system shall exclude retired or disposed assets from availability while retaining their history for authorized review and reporting. **Acceptance:** Disposed asset is unavailable but its history remains viewable.

4.7 AI-Assisted Asset Classification

4.7.1 Description and Priority

Priority: High. Suggest metadata for human confirmation (Phase 1, Section 4.6).

4.7.2 Stimulus/Response Sequences

User supplies text/files → AI suggests metadata/uncertainty → user reviews → confirmed values and review history are saved. Failure permits manual completion. See UC-02.

4.7.3 Functional Requirements

FR-AI-CLASS-001 [Must] The system shall analyze supplied asset titles, descriptions, specifications, images, and documents to recommend categories, subcategories, standardized tags and descriptions, and

possible material types, and identify missing metadata. **Acceptance:** Representative test assets receive suggestions for each listed item.

FR-AI-CLASS-002 [Must] The system shall allow authorized users to accept, correct, or reject suggested metadata within the registration or editing workflow, and shall not overwrite confirmed information without user acceptance. **Acceptance:** Rejecting a suggestion leaves confirmed metadata unchanged; the suggestion and review decision remain in the AI audit record.

FR-AI-CLASS-003 [Must] The system shall identify uncertain or unsupported suggestions. **Acceptance:** Low-certainty test cases are visibly marked.

FR-AI-CLASS-004 [Must] If classification fails or cannot produce a usable suggestion, the system shall say that assistance is unavailable and allow manual entry without losing supplied information. **Acceptance:** With the AI service disabled, manual registration succeeds with no data lost.

4.8 AI Resource Matching, Ranking and Explanations

4.8.1 Description and Priority

Priority: High. Find, rank and explain semantic matches without authorizing transactions (Phase 1, Section 4.7).

4.8.2 Stimulus/Response Sequences

Active need/listing triggers matching → system checks constraints, scores and explains → user selects interest only. Failure offers conventional search; remaining need stays active. See UC-06.

4.8.3 Functional Requirements

FR-AI-MATCH-001 [Must] The system shall match departmental requests with available listings using semantic meaning rather than relying only on identical keywords. **Acceptance:** Synonym test cases (e.g., “display” and “monitor”) are matched.

FR-AI-MATCH-002 [Must] Matching shall consider category compatibility, functional purpose, technical specifications, quantity, asset condition, requested condition, location, urgency, required date, transfer cost, and repair cost. **Acceptance:** On prelabeled cases, each factor is used in suitability checking or scoring as specified by the documented scoring rules; mandatory incompatibility is blocked and missing mandatory data is flagged. See TBD-004.

FR-AI-MATCH-003 [Must] The system shall generate a compatibility score for each recommended asset-request combination. **Acceptance:** Every recommendation displays a score.

FR-AI-MATCH-004 [Must] The system shall rank suitable matches by functional suitability, urgency, transfer cost, repair requirements, waste-diversion benefit, and estimated carbon-emission reduction. **Acceptance:** Use prelabeled ranking cases and documented weights (TBD-004); scores and order agree with the selected rules, and sustainability benefits do not override mandatory suitability.

FR-AI-MATCH-005 [Must] Each recommendation shall include a readable explanation of the main factors that influenced it, including any uncertainty or missing information. **Acceptance:** Reviewers confirm that explanations agree with the underlying data.

FR-AI-MATCH-006 [Must] Listings with a confirmed incompatibility with a mandatory requirement shall not be presented as fully suitable matches. **Acceptance:** Incompatible test listings are excluded or labeled unsuitable.

FR-AI-MATCH-007 [Must] Missing mandatory information shall be shown as uncertain and shall not be treated as a confirmed match. **Acceptance:** Listings missing mandatory data are flagged as unconfirmed.

FR-AI-MATCH-008 [Must] The system shall allow users to accept or reject recommendations and retain the recommendation and the review outcome. **Acceptance:** Accept and reject decisions are stored with the recommendation.

FR-AI-MATCH-009 [Must] Selecting a recommendation shall not bypass reservation, availability, or approval rules. **Acceptance:** Accepting a match only creates an expression of interest.

FR-AI-MATCH-010 [Must] If no suitable match is found or the matching service is unavailable, the system shall explain the outcome and allow conventional search without closing the request. **Acceptance:** With AI disabled, the request stays open and keyword search works.

FR-AI-MATCH-011 [Must] The system shall notify requesters when a newly published listing matches an active request with unallocated outstanding need, including Reserved or Partially Fulfilled requests (Phase 1, Req. 53). **Acceptance:** A matching publication notifies the requester for remaining unallocated demand; closed, canceled and fully fulfilled requests receive no match notification.

4.9 Sustainable-Action Recommendations

4.9.1 Description and Priority

Priority: Medium. Advise on sustainable next actions for human decision (Phase 1, Section 4.8).

4.9.2 Stimulus/Response Sequences

Custodian requests advice → AI assesses records/uncertainty → user reviews/overrides → retain, transfer, maintenance or approved end-of-life workflow follows. See UC-10.

4.9.3 Functional Requirements

FR-AI-SUST-001 [Must] The system shall recommend retention, transfer, repair, refurbishment, donation, component recovery, recycling, or retirement, with disposal considered only as a final option. **Acceptance:** Test assets receive one of these actions; disposal is never preferred while a feasible alternative exists.

FR-AI-SUST-002 [Must] Recommendations shall consider recorded condition, maintenance history, departmental demand, transfer and repair costs, and potential financial and environmental benefits. **Acceptance:** Prelabeled cases verify that each recorded factor appears in the assessment; infeasible actions are rejected and missing essential inputs produce a review warning. The reviewer checks the explanation against the records.

FR-AI-SUST-003 [Must] The system shall explain each recommendation, identifying supporting information, assumptions, and uncertainties. When essential information is missing, it shall indicate the need for further review instead of a definitive recommendation. **Acceptance:** An asset with missing data returns “needs review”.

FR-AI-SUST-004 [Must] The system shall allow authorized users to accept, reject, or override a recommendation and record their decision and reason. Accepting a recommendation shall not execute the action or bypass its approvals. **Acceptance:** Acceptance records the decision and opens the applicable workflow. Retention leaves the current state unchanged; repair/refurbishment uses maintenance; end-of-life completion remains blocked until approval.

4.10 Natural Language Assistant

4.10.1 Description and Priority

Priority: Medium. Provide permitted natural-language guidance and request preparation (Phase 1, Section 4.9).

4.10.2 Stimulus/Response Sequences

User asks → assistant uses accessible records/policy → answers or prepares a draft → user reviews before normal submission. Transactions use authorized workflows; outage offers conventional pages. See UC-11.

4.10.3 Functional Requirements

FR-ASSIST-001 [Must] The assistant shall support natural-language resource search, request preparation, questions about asset history and request status, policy guidance, sustainability information, and navigation to system functions. **Acceptance:** A scripted set of questions in each category is answered.

FR-ASSIST-002 [Must] When preparing a request, the assistant shall let the user review and correct the interpreted details before submission through the standard request workflow. **Acceptance:** An assistant-prepared request opens in the normal form for review.

FR-ASSIST-003 [Must] The assistant shall base factual answers on records accessible to the user and policy information supplied to the system, and shall say when information is unavailable or insufficient. **Acceptance:** Questions about restricted or missing records do not reveal data and state that no answer is available.

FR-ASSIST-004 [Must] The assistant shall not approve, reserve, transfer, or dispose of assets; such actions shall remain in the normal workflows (BR-002). **Acceptance:** Requests to approve or transfer are declined with a link to the workflow.

FR-ASSIST-005 [Must] If the assistant is unavailable or cannot complete a task, the system shall allow the user to continue through conventional search, forms, and status pages. **Acceptance:** With the assistant disabled, all tasks remain possible.

4.11 Reporting, Notifications and Auditability

4.11.1 Description and Priority

Priority: Medium. Provide reports, AI summaries, notifications and auditability (Phase 1, Sections 4.10/4.11).

4.11.2 Stimulus/Response Sequences

User chooses permitted filters → system calculates totals/estimates → AI summarizes the same figures. Events create permitted notifications/audit records. Missing inputs are identified. See UC-12 and Section 6.2.

4.11.3 Functional Requirements

FR-REPORT-001 [Must] The system shall provide operational reports on available assets, outstanding requests, reservations, completed transfers, maintenance activity, ownership changes, and approvals, filterable by department, category, and reporting period. **Acceptance:** Report totals match the test data for each filter.

FR-REPORT-002 [Must] Reports shall distinguish completed activities from pending or canceled actions. **Acceptance:** Canceled test transfers are not counted as completed.

FR-REPORT-003 [Must] The system shall display the sustainability indicators defined in Section 6.2 by department, category, and reporting period. **Acceptance:** Dashboard counts, rates and estimates match independently worked examples, including zero denominators, missing inputs and duplicate activity records.

FR-REPORT-004 [Must] The system shall generate AI-assisted summaries of asset utilization, reuse activity, maintenance trends, departmental demand, estimated savings, and sustainability outcomes. **Acceptance:** A summary is produced for a selected period.

FR-REPORT-005 [Must] Numerical values in AI-generated reports shall agree with the underlying calculations. Reports shall identify their reporting period, data basis, and model version, and distinguish recorded outcomes from estimates. **Acceptance:** Every number in a generated report is checked against the source figures.

FR-REPORT-006 [Must] The system shall provide in-application notifications about new matches, pending approvals, request and reservation status changes, transfer arrangements, ownership changes, and maintenance updates. **Acceptance:** Each event type produces a notification for the right user.

FR-REPORT-007 [Must] Each notification shall identify the event, the related asset or request, and any required action, and link to the corresponding record or permitted workflow step. **Acceptance:** Opening a notification leads to the correct record.

FR-REPORT-008 [Must] Notification visibility shall respect the recipient's permissions, and failure to deliver a notification shall not alter the underlying record. **Acceptance:** A simulated delivery failure leaves the record unchanged.

FR-REPORT-009 [Must] The system shall maintain an audit trail of approvals, ownership and custody changes, and decisions on AI recommendations, which authorized users can review. **Acceptance:** Account, approval, formal custody, physical handover and AI review events are visible to authorized reviewers with actor, time and before/after values where applicable.

5 Other Nonfunctional Requirements

5.1 Performance Requirements

NFR-PERF-001 Ordinary filtered searches shall complete within 2 seconds for at least 95% of measured requests with five simultaneous user sessions (Phase 1, Req. 56). The test shall declare dataset size, query mix, network, browser and hosting environment. The Phase 2 planning dataset is approximately 5,000 asset records, including 500 published listings. **Verification:** Load test with recorded samples and percentile calculation on the declared synthetic dataset.

NFR-PERF-002 Longer AI operations shall display a processing status while the user waits (Phase 1, Req. 61). Target: Status shown for every AI operation that is not immediate. **Verification:** System test

NFR-PERF-003 Response-time targets for AI-assisted operations shall be set after prototype technical checks; see TBD-005. **Verification:** Measured during implementation

5.2 Safety Requirements

NFR-SAFE-001 The system shall not automatically authorize physical transfers, disposal, or recycling. **Verification:** Workflow test

NFR-SAFE-002 Tests involving assets recorded as unfit or incompatible with mandatory requirements shall confirm that ordinary-use reservation or release is prevented. AI assistance shall not bypass these checks (Phase 1, Req. 57). **Verification:** Workflow test with unfit and incompatible test assets

5.3 Security Requirements

NFR-SEC-001 The system shall enforce role-based authorization for every protected request, including direct access attempts and assistant interactions (Phase 1, Req. 58). **Verification:** Security test of allowed and denied actions for each role

NFR-SEC-002 Passwords shall not be stored in plaintext, and network access shall use encrypted connections (Phase 1, Req. 59). **Verification:** Configuration review and test

NFR-SEC-003 Credentials and sensitive configuration shall be excluded from the shared source repository and shall not be exposed to users' browsers (Phase 1, Req. 59; Phase 2, Section 4.1). **Verification:** Repository and code review

NFR-SEC-004 Access to financial and operational information shall be limited to roles authorized for it (project description; Section 2.3). **Verification:** Security test

NFR-PRIV-001 Privacy The system shall not send confidential data to an AI service without authorization. **Verification:** Review/test

NFR-PRIV-002 Privacy Only the information needed for a task shall be sent to an AI service, and the prototype shall use synthetic data only (Phase 2, Sections 4.2 and 6.2). **Verification:** Review of AI request content

5.4 Software Quality Attributes

NFR-QUAL-001 Usability Core publication, request, approval, and receipt tasks shall be completable using on-screen guidance without developer intervention (Phase 1, Req. 60). **Verification:** User-acceptance test with reviewers who did not build the task (Phase 1, Section 6)

NFR-QUAL-002 Usability Forms shall support keyboard operation, identify errors clearly, and preserve entered information when validation fails (Phase 1, Req. 60). **Verification:** Usability test

NFR-QUAL-003 Reliability Confirmed records shall survive an application restart, and failed operations shall not leave partially applied reservations or transfers (Phase 1, Req. 61). **Verification:** System test

NFR-QUAL-004 Recoverability Database records and uploaded files shall be capable of being backed up and restored (Phase 2, Section 4.4). **Verification:** Backup and restore test

NFR-QUAL-005 Maintainability Setup instructions, configuration, and component interfaces shall be documented so another team member can run and maintain the prototype; AI integrations shall remain separable from the rules controlling authorization and asset transactions (Phase 1, Req. 62). **Verification:** Peer review; setup by a second member

NFR-QUAL-006 Resource efficiency Conventional operations shall not invoke AI merely to enforce fixed business rules, and AI-assisted operations shall record call counts, processing time, and estimated service cost where pricing information is available (Phase 1, Req. 63). **Verification:** Log review

NFR-AI-001 Explainability AI recommendations shall display their main influencing factors and allow human override. **Verification:** AI evaluation

5.5 Business Rules

ID	Business rule	Rationale
BR-001	Only an authorized department representative or asset custodian may publish or withdraw an asset of their department (Phase 1, Req. 7).	Accountability for records
BR-002	AI recommendations shall not automatically approve reservations, transfers, disposal, donation, or recycling (Phase 1, Section 2).	Human oversight
BR-003	A requester shall not approve their own transfer request unless explicitly authorized.	Separation of duties
BR-004	Publication does not require central pre-approval, and publication alone does not approve a transfer (Phase 1, Reqs. 7 and 10).	Low-effort listing without weakening control
BR-005	Competing eligible requests are normally considered in submission order; exceptions based on urgency, suitability, or timing require a recorded reason (Phase 1, Section 5.5).	Fairness and traceability
BR-006	Related approvals may be combined only within the representative's authority (Phase 1, Section 5.5).	Delegated authority
BR-007	A transfer is complete only after the receiving representative accepts receipt of the approved quantity. Formal departmental custody changes at completion; actual physical holder and location are tracked during movement and discrepancies (Phase 1, Req. 22).	Custody accuracy
BR-008	Donation, component recovery, recycling, and disposal require authorized approval before being recorded as completed (Phase 1, Req. 27; Phase 2, Section 2.1).	Delegated authority

ID	Business rule	Rationale
BR-009	Lack of demand or elapsed storage time does not automatically authorize disposal (Phase 1, Section 5.5).	Reuse before disposal
BR-010	Assets recorded as unfit remain unavailable for ordinary-use reservation until an authorized condition assessment establishes suitability. Rejecting an end-of-life proposal does not make an unfit asset usable (Phase 1, Req. 26).	Safety
BR-011	Technical administration privileges do not grant business approval authority (Phase 1, Req. 4; Phase 2, Section 2.1).	Separation of duties

The role scopes in Section 2.3 implement these rules. Detailed delegation, retention and exception policies remain TBD-006; an AI suggestion, technical role or expired storage period cannot supply missing approval authority.

6 Other Requirements

6.1 AI Requirements and Evaluation

These controls apply to every AI feature in Section 4. AI remains advisory; people retain authority for consequential actions. Confidence labels shall not imply calibrated probabilities unless the method has been validated (TBD-004).

AI-REQ-001 The system shall record the AI model/version used for each AI output; if the provider does not expose an exact revision, it shall retain the exposed model identifier, application configuration/version and time, marking that limitation. **Verification:** Test

AI-REQ-002 The system shall provide a non-AI fallback for search and approval workflows. **Verification:** Test with AI services disabled

AI-REQ-003 AI output shall not approve, reserve, release, transfer, change ownership of, or record the donation, recycling, or disposal of an asset; these actions shall require an authorized user (Phase 1, Section 2; BR-002). **Verification:** Workflow test

AI-REQ-004 The system shall distinguish AI suggestions from confirmed information and show their confidence or uncertainty (project description, Phase 3; Phase 1, Reqs. 30-31 and 36). **Verification:** Test

AI-REQ-005 The system shall record each user's acceptance, correction, rejection, or override of an AI output (Phase 1, Reqs. 31, 37, and 42). **Verification:** Test

AI-REQ-006 The application shall check required fields, suitability, available quantities, and permissions before acting on AI output (Phase 2, Section 4.3). **Verification:** Test with invalid AI output

AI-REQ-007 AI matching shall be compared with keyword and category search on the same requests and listings, reporting suitable matches within the first three results and unsuitable suggestions. Before testing, reviewers shall label suitability against the request's mandatory constraints, independently of the AI output. Classification shall be checked against supplied metadata. Sustainable actions, assistant answers and report summaries shall be checked against records, permitted workflows and calculation results. **Verification:** Evaluation report with fixed cases, expected outcomes, both result sets, model/configuration, uncertainty and limitations; no unsupported accuracy claim.

AI-REQ-008 Testing shall check whether recommendations favor particular departments without a justified reason (Phase 2, Section 6.2). **Verification:** AI evaluation report

6.2 Sustainability Measurement Requirements

Calculations use completed outcomes and declared inputs. Synthetic-data reports demonstrate calculation correctness, not actual campus-wide savings. Every report identifies its period, scope, unit and missing information. Quantities count physical units or measured mass, never a mixture of units or a count of database rows. Aggregation of unlike asset categories must show category totals or use a justified common unit.

SUS-001 Avoided purchases Count accepted transferred units that fulfil a need explicitly identified by the requester or test scenario as otherwise requiring purchase. Gross avoided-purchase value = eligible units × stated replacement unit value. Do not assume every reuse prevented a purchase. **Verify** with worked cases including no purchase basis and partial receipt.

SUS-002 Waste diversion For a declared disposal-risk cohort and period, count units or mass with a completed reuse, repair, donation, component-recovery or recycling outcome that avoids the disposal baseline. Diversion rate = diverted quantity ÷ (diverted + disposed quantity) × 100 for completed outcomes in that same cohort/period. Pending outcomes are reported separately and excluded from this denominator. Use one unit consistently; a zero denominator is N/A. Count each quantity once, even if repaired then transferred. **Verify** mixed, duplicate and pending outcomes.

SUS-003 Estimated carbon reduction Estimate avoided new-purchase emissions for eligible reuse units using stated replacement/emission factors; subtract relevant repair/transfer emissions where data exists and the selected method requires them. Declare boundary, units, factor sources, omissions and uncertainty. Missing essential factors produce unavailable results. Factor sources and values remain TBD-002. **Verify** against independently calculated examples; do not label an estimate as measured impact.

SUS-004 Net financial savings Net savings = gross avoided-purchase value (SUS-001) minus relevant repair and handling/transport costs for that fulfillment. Allocate shared costs explicitly and once. Do not subtract unrelated costs or count the same avoided purchase twice. **Verify** single and partial-transfer examples, including negative savings.

SUS-005 Reuse and repair activity Show completed transfer-event counts separately from accepted reused-unit counts, and completed repair-event counts separately from units restored to service. For a selected listing cohort, reuse rate = distinct eligible units whose internal transfer receipt is accepted during the reporting period ÷ distinct eligible units available for reuse during that period × 100. State the cohort and count each unit once; zero denominator is N/A. Repair does not count as reuse until a transfer is completed. **Verify** rates and counts against labeled activity data.

SUS-006 Asset life extension Estimate added service life through reuse/repair using declared assumptions and a stated baseline. The metric remains in prototype scope; its method is TBD-003. Display unavailable where the method or required inputs are absent, and keep service-life estimates distinct from measured outcomes. **Verify** the selected method and examples after design.

SUS-007 Calculation rules Retain inputs, methods, units, factors, sources and assumptions with results. Identify missing inputs rather than inventing values. Deduplicate unit/outcome cohorts and avoided purchases; retain links to supporting records. Distinguish event counts, physical quantities, rates and estimates. **Verify** missing-input, duplicate and denominator cases independently.

6.3 Data Documentation and Delivery Requirements

DD-001 The prototype shall include a reproducible synthetic starting dataset with consistent links among users, departments, assets, batch quantities, requests, allocations, approvals, transfers, maintenance and audit records. It shall support independently reproducible report and AI-evaluation examples, including partial allocations, returned/unfit items, missing inputs and competing approvals. **Verification:** reload the dataset and reproduce the declared test outcomes.

DD-002 Setup and user documentation shall accompany the prototype as specified in Section 2.6 and NFR-QUAL-005. Dependency licences and hosted-service terms shall be respected; example photographs/documents shall be team-created or used with permission (Phase 2, Section 6.1). **Verification:** delivery checklist and licence/asset-source review. Any future use of real personal data requires institutional and applicable data-protection review; it is outside this synthetic-data submission.

Appendix A: Glossary

Term	Definition
AI	Artificial intelligence. In this system, AI suggests and explains; authorized people decide.
Asset	A campus resource recorded in the system, either an individually tracked item or a quantity-based batch.
Asset Custodian	The person responsible for the day-to-day care, records, and availability of an asset.
Avoided purchase	A purchase that did not occur because an internal transfer fulfilled the need.
Circular Campus	A campus that keeps resources in use through reuse, repair, refurbishment, and recycling before disposal.
Compatibility score	A score showing how well a listed asset suits a departmental request.
Component recovery	Disassembling an asset to reuse its working parts.
End-of-life action	Donation, component recovery, recycling, retirement, or disposal of an asset.
Expression of interest	A requester’s request for a listed asset; it does not reserve any quantity.
LLM	Large language model; the type of AI used by the conversational assistant and report summaries.
Semantic Matching	Matching requests with listings by meaning rather than by identical keywords.
Synthetic data	Realistic but artificial records used to build and test the prototype without real university data.
Waste diversion	Keeping assets or materials from disposal through reuse, repair, donation, component recovery, or recycling.
Formal departmental custody	Accountable departmental/custodian assignment; a transfer changes this only after accepted receipt.
Physical holder and location	The person or unit holding an asset and its actual location during handover, transit, discrepancy or return.
Actively allocated quantity	Quantity tied to confirmed reservations or unresolved transfers and not yet accepted or reconciled as returned/canceled.
Unallocated outstanding quantity	Requested quantity minus accepted fulfillment and active allocations; the quantity eligible for a further reservation.
Batch portion	An allocated quantity of a batch with its own condition/status tracking; other portions may remain available.

Appendix B: Analysis Models

These Phase 3 models specify participation and controlled workflows. Sign-in and role authorization apply throughout; AI cannot approve transactions. Section 2.3 controls exact permissions. Detailed design models follow in later phases.

B.1 Use Case Diagram

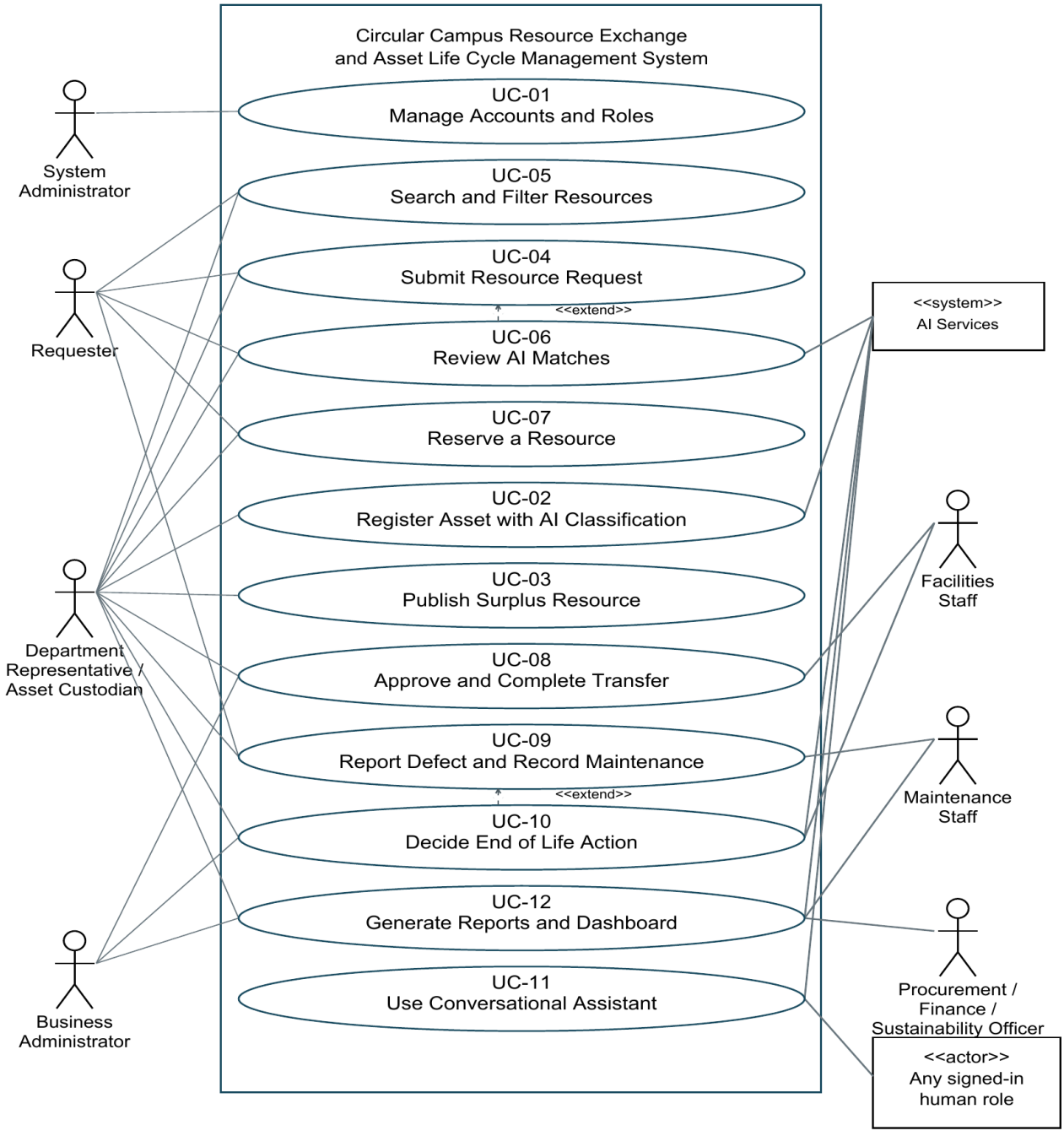


Figure 1 Use case diagram with system boundary and supporting actors

UC-06 extends UC-04 when matching is used; UC-10 extends UC-09 after an unfit outcome and can also start independently. UC-11 serves every signed-in human role. Representatives may perform requester actions. Exact permissions remain in Section 2.3.

B.2 Asset Life Cycle State Model

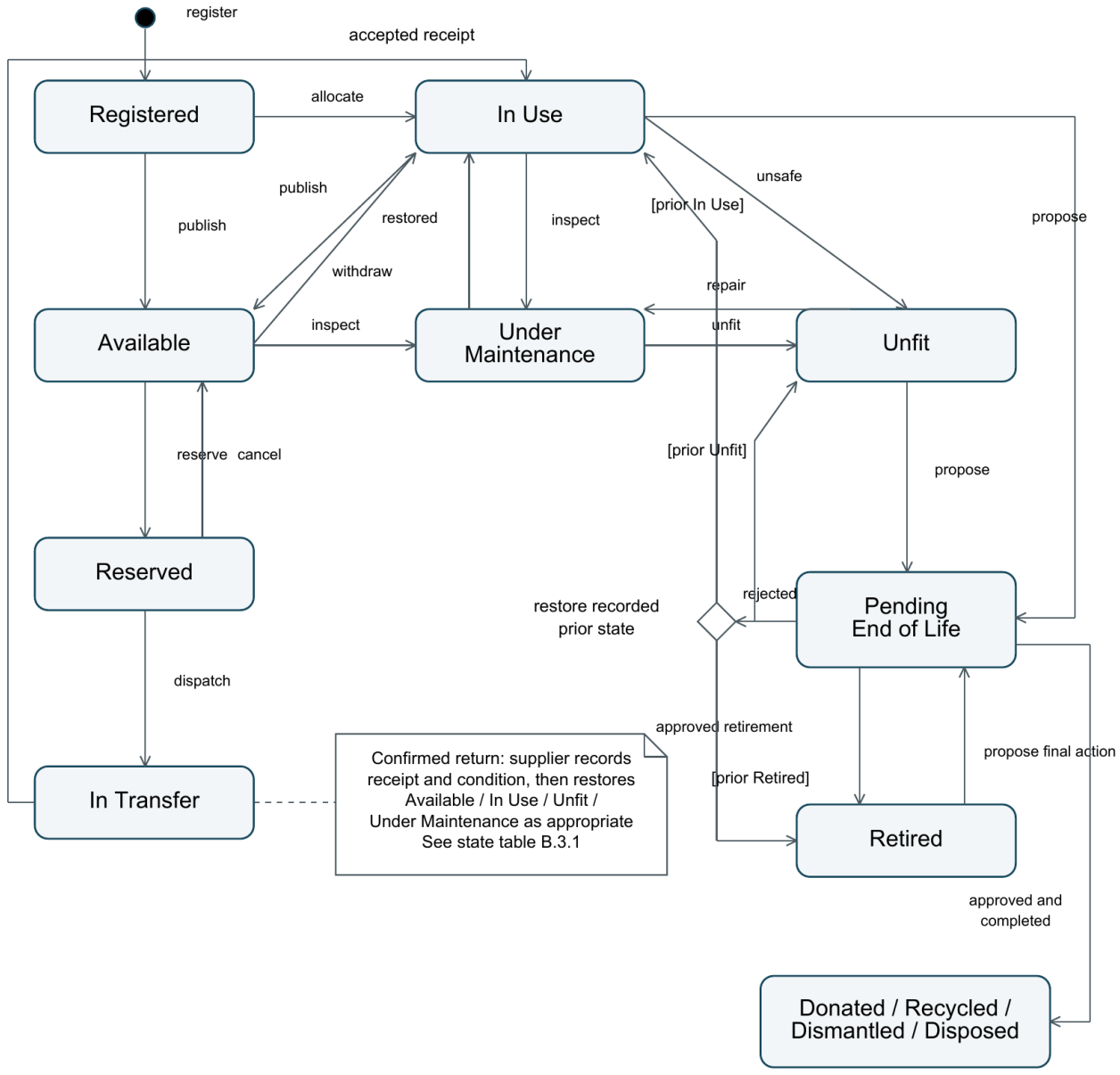


Figure 2 Asset state model with guarded receipt return and prior-state rejection outcomes

This overview groups final end-of-life outcomes and shows the principal service/reuse paths. The complete permitted transitions, including condition failures in Available/Reserved, restored availability after maintenance and coordinated return quantities, are specified in B.3.1. A rejection restores the pre-proposal state; it does not certify safety or restore availability by itself.

B.3 Workflow States and Approval Stages

States apply to an individually tracked asset or a defined batch portion, not automatically to every unit in a batch. Different portions may occupy different states. A batch of 10 with 3 Reserved therefore

retains 7 Available. Preserve quantity totals and allocation links after every transition. Transitions not listed are invalid. Final end-of-life units remain in history but are excluded from stock. Tables and diagrams use the same transition guards. Role scopes, safety gates and allocation controls in Sections 4 and 5.5 and the use cases remain binding: unsafe/disputed stock stays unavailable, cancellation does not clear safety flags, and end-of-life proposals require no active allocation.

B.3.1 Asset States

State	Permitted next states and conditions	Responsible role or control
Registered	In Use (allocated); Available (published).	Department Representative / Asset Custodian.
In Use	Available (published); Under Maintenance; Unfit; Pending End-of-Life.	Custodian; Maintenance Staff assess condition.
Available	Reserved; In Use (withdrawn); Under Maintenance; Unfit when unsafe condition is confirmed.	Supplying representative; authorized condition assessor.
Reserved	In Transfer (dispatch); Available (canceled before dispatch); Under Maintenance or Unfit after a condition failure, with allocation released and need restored.	Supplying representative / Facilities Staff; condition assessor.
In Transfer	In Use at receiver (accepted receipt); Available or In Use at supplier (confirmed suitable return); Unfit or Under Maintenance (confirmed unsafe/inspection-required return).	Receiver accepts; supplier confirms return/condition.
Under Maintenance	In Use or Available (safe restoration to prior service/listing context); Unfit (unsuitable outcome).	Maintenance Staff; authorized condition update.
Unfit	Under Maintenance; Pending End-of-Life.	Custodian / Maintenance Staff.
Pending End-of-Life	Retired after authorized retirement; Donated/Recycled/Dismantled/Disposed after authorized completed action; restore prior In Use, Unfit or Retired state if rejected.	Business Administrator approves; staff record completion.
Retired	Pending End-of-Life for a proposed final action.	Custodian.
Donated / Recycled / Dismantled / Disposed	No further transition; retain history.	Final; history retained.

B.3.2 Request States

Requested, accepted, active-allocation and unallocated-outstanding quantities are maintained separately. Reserved means active allocation exists without accepted fulfillment; Partially Fulfilled means some but not all units were accepted, with or without further active allocation. Either state permits further reservation of unallocated outstanding need. Returned quantity is not accepted fulfillment. Final closure does not imply that all requested units were received.

State	Permitted progression	Role or condition
Open	Reserved; Canceled.	Requester submits/cancels; supplier approves allocation.
Reserved	Reserved with additional eligible allocations; Partially Fulfilled; Fulfilled; Open if all allocations released; Canceled before dispatch.	Supplier allocates; receiver accepts receipt. Reconcile dispatch before cancellation.
Partially Fulfilled	Partially Fulfilled with further allocations/receipts; Fulfilled; Closed.	Receiver accepts; requester may close after allocations are reconciled.
Fulfilled / Closed / Canceled	No further allocations. History retained.	All units accepted for Fulfilled; requester explicitly closes unmet need.

B.3.3 Transfer States

State	Permitted progression	Responsible role or control
Pending Approval	Approved; Canceled.	Supplying representative; rejection reconciles allocation.
Approved	Scheduled; Canceled before dispatch.	Facilities Staff / supplying representative.
Scheduled	In Transit on dispatch; Canceled before dispatch.	Facilities records physical holder/location.
In Transit	Completed after accepted receipt; Disputed after discrepancy.	Receiving representative.
Disputed	Completed after reconciled accepted receipt; Returned after confirmed return receipt and condition.	Representatives; Business Administrator handles escalation.
Completed / Returned / Canceled	No further transfer transition; history retained.	Update stock, allocation and request together; returned stock follows condition.

B.3.4 Maintenance States

State	Permitted progression	Responsible role or control
Reported	Under Inspection; Canceled.	Requester/custodian reports; Maintenance Staff accepts.
Under Inspection	Under Repair; Completed if no repair needed.	Maintenance Staff.
Under Repair	Completed with restored or unsuitable outcome.	Maintenance Staff.
Completed	Closed after authorized condition/state update.	Custodian / Maintenance Staff update condition.

State	Permitted progression	Responsible role or control
Closed / Canceled	No further maintenance transition.	Retain history and safety assessment.

B.4 Use Case Descriptions

Each use case includes its goal, actors, conditions, trigger, success/alternative/exception paths and outcomes. “Representative” means the authorized departmental representative/custodian. Sections 2.3 and 5.5 constrain all actions.

UC-01 Manage User Accounts and Roles

Description: Create, edit or deactivate an account and its authorized access.

Actors: Primary: System Administrator. Secondary: affected user.

Preconditions: Administrator signed in; department exists.

Trigger: Open Add user or an existing account.

Main Success Flow

1. Administrator enters identity, department and assigned roles.
2. System checks required data and duplicate email; administrator confirms.
3. System saves account/changes and audits actor, time and before/after values.
4. User signs in with the assigned permissions.

Alternative Flows: Edit membership/roles; deactivate while retaining historical records. Role changes apply at the next sign-in; the resulting session must enforce the new scope.

Exception Flows: Duplicate email or missing information blocks the save and preserves valid entered data.

Postconditions: Account is saved or deactivated, with audit history; failed validation creates no new account.

Rules and Requirements: BR-011; FR-USER-001 to 005.

UC-02 Register Asset with AI Classification

Description: Create an asset record with human-reviewed classification suggestions.

Actors: Primary: representative. Secondary: AI Services.

Preconditions: Permission to register the department’s assets.

Trigger: Select Register asset.

Main Success Flow

1. User enters title, description/specifications and uploads relevant files.
2. On request, AI suggests category/subcategory, tags, description, materials and missing information, marking uncertainty.
3. User accepts, corrects or rejects suggestions and completes remaining fields.
4. System validates, assigns an ID and saves confirmed data; it records suggestions, decisions and model/version.

Alternative Flows: Register manually without AI; after saving, publish through UC-03.

Exception Flows: AI failure permits manual completion without lost input. Invalid fields block save and retain entered data.

Postconditions: Saved record is Registered, or Available after successful UC-03 publication. Rejected suggestions remain only in the review log, not confirmed metadata.

Rules and Requirements: AI-REQ-004/005; FR-ASSET-001/002 and FR-AI-CLASS-001 to 004.

UC-03 Publish Surplus Resource

Description: Make eligible departmental surplus discoverable.

Actors: Primary: representative.

Preconditions: Asset exists in a publishable state; permitted quantity has no conflicting allocation.

Trigger: Select Publish on an asset.

Main Success Flow

1. System pre-fills saved details; user completes availability and confirms quantity/condition.
2. System validates publication fields and authority.
3. System publishes eligible quantities as Available and records history.

Alternative Flows: Correct saved details; withdraw an unallocated listing to In Use.

Exception Flows: Missing/invalid fields block publication; a reservation/transfer blocks withdrawal of the affected quantity. Unknown condition is visibly marked, not certified suitable.

Postconditions: Eligible quantities appear in search; publication supplies no transfer approval. Failure leaves the previous state unchanged.

Rules and Requirements: BR-001/004/010; FR-ASSET-003 to 008.

UC-04 Submit Resource Request

Description: Record a departmental need once and offer suitable resources.

Actors: Primary: requester. Secondary: AI Services through UC-06.

Preconditions: Requester belongs to a department.

Trigger: Open New request or an assistant-prepared request.

Main Success Flow

1. Requester enters category, purpose, specifications, quantity, condition, urgency, location and required date.
2. System validates; requester confirms submission.
3. System saves an Open request and displays quantities, next action and role.
4. When available, matching runs at the UC-06 extension point.

Alternative Flows: Review/correct a pre-filled assistant request; use conventional search (UC-05).

Exception Flows: Missing fields, non-positive quantity or invalid date block submission while retaining input. AI failure leaves the saved request Open and offers conventional search.

Postconditions: An Open request is visible to the requester and authorized representatives; no asset is reserved.

Rules and Requirements: BR-005; FR-REQUEST-001 to 003/005 and UC-06.

UC-05 Search and Filter Resources

Description: Find published eligible resources through conventional search.

Actors: Primary: requester or another permitted user.

Preconditions: Access to marketplace.

Trigger: Enter keyword or filters.

Main Success Flow

1. User supplies keywords/category and optional condition, quantity, location, age/value and state filters.
2. System returns matching listings with availability/condition.
3. User reviews details and may express interest linked to a request.

Alternative Flows: Change/combine filters; search while AI is unavailable.

Exception Flows: No result shows a message and leaves any active need open. Retired/disposed resources are excluded from available results.

Postconditions: Results reflect permitted current stock; searching alone reserves no quantity.

Rules and Requirements: BR-010; FR-SEARCH-001 to 004.

UC-06 Review AI Match Recommendations

Description: Review scored, explained matches for remaining departmental need.

Actors: Primary: requester. Secondary: AI Services.

Preconditions: Active Open, Reserved or Partially Fulfilled request with unallocated outstanding need.

Trigger: Submit/edit a request, publish a matching listing or open recommendations.

Main Success Flow

1. System compares meaning and recorded suitability factors with available listings.
2. It rejects/labels mandatory incompatibilities and flags missing mandatory data.
3. It scores and ranks suitable candidates with explanations, uncertainty and model details.
4. Requester accepts/rejects; system logs the review. Selection opens linked interest through UC-07.

Alternative Flows: Accept several partial candidates only within remaining need; reject with optional reason.

Exception Flows: No suitable match or unavailable AI offers conventional search without closing remaining need. Changed availability is rechecked during reservation.

Postconditions: Recommendations/reviews are retained; no stock is reserved or approved by AI selection.

Rules and Requirements: BR-002/010; FR-AI-MATCH-001 to 011.

UC-07 Reserve a Resource

Description: Reserve approved available quantity without over-allocation.

Actors: Primary: requester. Secondary: supplying representative.

Preconditions: Listing has suitable Available quantity; request is active with unallocated outstanding need.

Trigger: Select Request this item or accept a match.

Main Success Flow

1. Requester selects quantity and links the request; system records interest only and notifies supplier.
2. Supplier reviews competing interests in order and approves an eligible quantity.
3. System atomically checks stock, mandatory suitability and remaining unallocated demand.
4. System confirms allocation, reduces available quantity, updates status and notifies parties; it creates a transfer record.

Alternative Flows: Approve a later interest with recorded justification; approve a partial quantity; add further eligible reservations in Reserved/Partially Fulfilled. Cancel before dispatch to release allocation.

Exception Flows: Changed stock, excess demand or unsafe/incompatible condition blocks confirmation. Rejection stores the reason and creates no allocation.

Postconditions: Confirmed allocations never exceed stock or demand. Remaining unmet need stays active; pending interests reserve nothing.

Rules and Requirements: BR-003/005/010; FR-REQUEST-003 to 007.

UC-08 Approve and Complete Transfer

Description: Record authorized release, physical handover and accepted receipt or reconciled return.

Actors: Primary: supplying representative. Secondary: Facilities Staff, receiving representative, Business Administrator for escalations.

Preconditions: Confirmed reservation; required suitability established; item not Unfit.

Trigger: Reservation creates a transfer record.

Main Success Flow

1. System links assets/quantities, departments, destination and arrangements.
2. Supplying representative approves release; Facilities Staff schedule and record dispatch with actual physical holder/location.
3. Receiving representative records quantity and condition and accepts approved receipt.
4. System completes transfer and updates formal ownership where authorized, departmental custody, destination assignment, availability, request fulfillment, history, costs and notifications together.

Alternative Flows: Combine authorized reservation/release decisions; receiving department collects.

Exception Flows: Missing approval or suitability blocks release. A discrepancy keeps the transfer Disputed, allocated and unavailable with actual tracking. Reconcile accepted receipt to complete, or confirm return receipt/condition at supplier to mark Returned and restore unmet need. Suitable returned units regain the appropriate supplying state; unsafe units remain Unfit/under inspection. Pre-dispatch cancellation releases stock.

Postconditions: Completed receipt changes formal assignment. Disputed stock remains open. Confirmed return or cancellation reconciles allocation and condition without treating the need as fulfilled.

Rules and Requirements: BR-006/007/010; FR-TRANSFER-001 to 008.

UC-09 Report Defect and Record Maintenance

Description: Record a reported fault, inspection/service, costs and assessed condition.

Actors: Primary: requester or custodian reports; Maintenance Staff perform recorded service. Secondary: representative notified.

Preconditions: Accessible asset exists and is not in a final end-of-life state.

Trigger: Select Report issue.

Main Success Flow

1. User describes defect and requests inspection/repair; system links a Reported work order.
2. Maintenance inspect and record findings.
3. Staff record repair/refurbishment, costs and outcome.
4. Authorized condition update records safe restoration or Unfit, preserves/reconciles allocations and updates history.

Alternative Flows: No fault found completes inspection with supporting assessment. An unrepairable outcome permits UC-10 next-action review. Restore Available when the prior eligible listing context applies.

Exception Flows: Unsafe assessment blocks ordinary-use reservation/release immediately; missing required findings block completion. Cancellation of a work order does not clear an unsafe assessment.

Postconditions: Maintenance history and costs are retained; return to service requires assessed suitability, not merely completion of a work order.

Rules and Requirements: BR-010; FR-MAINT-001 to 004/006.

UC-10 Decide End of Life Action

Description: Review sustainable next actions and obtain authorization for a selected end-of-life outcome.

Actors: Primary: representative/custodian. Secondary: AI Services, Business Administrator, Maintenance/Facilities Staff as applicable.

Preconditions: Asset is unfit, retired or no longer needed; no active reservation/transfer on affected quantity.

Trigger: Select Review next action, independently or after UC-09.

Main Success Flow

1. System assembles condition/history, demand, costs and available benefit estimates.
2. AI recommends an action with support, assumptions and uncertainty; disposal is a final option.
3. Custodian reviews and records acceptance/override with reason and model details.
4. Retention ends review with the current safe state unchanged; transfer enters UC-03/08; repair/refurbishment enters UC-09.
5. For donation, component recovery, recycling, retirement or disposal, custodian submits a justified proposal; system records prior state and Pending End-of-Life.
6. Business Administrator approves; staff perform/record the authorized action; system updates final state, history and supported reports.

Alternative Flows: Missing information requires review or a reasoned manual choice. Rejected proposal records the reason and restores the prior appropriate In Use, Unfit or Retired state; Unfit remains unavailable.

Exception Flows: AI outage allows manual decision. Completion without approval is blocked. Active allocation prevents an end-of-life proposal.

Postconditions: Decision and approvals remain traceable. Retention/repair/transfer do not fall through into disposal; rejection never certifies safety.

Rules and Requirements: BR-002/008/009/010; FR-AI-SUST-001 to 004 and FR-MAINT-004/005.

UC-11 Use Conversational Assistant

Description: Obtain permitted natural-language guidance or prepare a request for review.

Actors: Primary: any signed-in human role. Secondary: AI Services.

Preconditions: User authorized for requested records/functions; supplied policy available where needed.

Trigger: Ask the assistant a question.

Main Success Flow

1. Assistant interprets the goal and retrieves only permitted relevant records/policy.
2. It answers with supported facts and uncertainty, or prepares editable request details.
3. User reviews/corrects any draft and submits through UC-04; transaction actions remain in standard workflows.

Alternative Flows: Open linked search, history/status or policy guidance; use conventional pages.

Exception Flows: Missing/restricted data gives no unsupported answer and reveals no restricted content. Requests to approve/reserve/transfer/dispose are directed to permitted workflows. AI outage offers the normal forms/pages.

Postconditions: An answer or editable draft is shown; assistant output itself changes no asset transaction.

Rules and Requirements: BR-002/011; FR-ASSIST-001 to 005.

UC-12 Generate Reports and Sustainability Dashboard

Description: Review role-permitted activity and reproducible financial/environmental estimates.

Actors: Primary: Procurement, Finance or Sustainability Officer, Business Administrator, representative or Maintenance Staff within report scope. Secondary: AI Services.

Preconditions: Report permission and selected dataset/period.

Trigger: Open reporting and choose filters.

Main Success Flow

1. User selects department/category/period within permitted scope.
2. System calculates completed activity, pending/canceled distinctions and Section 6.2 metrics from retained inputs.
3. It displays totals, rates and labeled estimates; AI summary uses the same figures.
4. User reviews period, units, source basis and uncertainty with supporting records.

Alternative Flows: Show fixed reports when AI is unavailable; select a narrower period/category.

Exception Flows: Unauthorized scope is rejected. Missing essential factors yield unavailable estimates; zero denominators yield N/A. Duplicate outcomes are excluded and unsupported AI numbers are not presented as verified values.

Postconditions: Reports agree with independently computed examples; reviewing a report changes no business record.

Rules and Requirements: BR-011; FR-REPORT-001 to 005, SUS-001 to 007 and AI-REQ-007.

Appendix C: To Be Determined List

These are design or policy decisions retained from the previous version; they do not remove the related functions from prototype scope. Group 6 owns each item. Close an item by recording the decision, its source/approval where required and the affected requirement IDs.

ID	Open item	Target decision
TBD-001	Confirm with the lab instructors that the technology assessed in Phase 2 is acceptable, given that the Phase 3 lab slides refer to a prototype in Python, Java, or C++.	Before Phase 4-5 submission
TBD-002	Source and values of the emission factors used for estimated carbon reduction (SUS-003).	Phase 4-5
TBD-003	Method for estimating asset-life extension (SUS-006).	Phase 4-5
TBD-004	How matching factors are weighted and how compatibility scores and confidence are presented (FR-AI-MATCH-003 to 005; AI-REQ-004).	Phase 4-5
TBD-005	Response-time targets for AI-assisted operations (NFR-PERF-003).	During Phase 4 technical checks, before implementation performance claims.
TBD-006	Detailed delegation, retention, and exception policies (Phase 1, Section 5.5).	Phase 4-5